Model Research Document

Project: AI Workforce Displacement & Employment Impact Analysis

This project analyzes the impact of Artificial Intelligence on workforce displacement using a structured dataset. Machine Learning (ML) models can identify patterns, classify jobs at risk, and predict workforce trends. AI models can further generate insights, summaries, and support decision-making.

Dataset Source: AI Workforce Displacement Global Dataset (2020–2026)

Models that can be implemented on this dataset

The dataset contains structured numerical and categorical features. Therefore, the following models can be implemented for prediction, classification, and workforce analytics:

Model	Can be Implemented?	Use Case	Advantages	Expected Performance
Logistic Regression	Yes	Binary classification	Simple and interpretable	82–87% Accuracy
Decision Tree	Yes	Workforce classification	Easy to explain	84–89% Accuracy
Random Forest	Yes	Displacement prediction	High accuracy, less overfitting	92–96% Accuracy
XGBoost	Yes	AI workforce prediction	Excellent performance	95–98% Accuracy
LightGBM	Yes	Workforce forecasting	Fast boosting algorithm	93–96% Accuracy
Support Vector Machine (SVM)	Yes	Classification	Works well on structured data	88–92% Accuracy
K-Nearest Neighbors (KNN)	Yes	Similarity prediction	Simple algorithm	84–89% Accuracy
Artificial Neural Network (ANN)	Yes	Complex pattern learning	Captures non-linear relationships	90–95% Accuracy
Linear Regression	Yes	Continuous value prediction	Simple baseline regression	R^2 depends on target
CatBoost	Yes	Categorical data prediction	Handles categorical features well	93–97% Accuracy
BERT	Optional	Text/skill analysis	NLP model	F1 Score: 91–94%

Llama 3	Optional	Insight generation & reporting	LLM for summaries and Q&A	Depends on benchmark

Conclusion

Among all the models, XGBoost is the most suitable for this dataset because it performs exceptionally well on structured tabular data and provides high prediction accuracy while handling missing values and reducing overfitting. Random Forest and LightGBM are also strong alternatives. BERT and Llama 3 should be considered only if the project includes text analysis, report generation, or an AI assistant.